Lemma A: Bahadur-Rao Theorem – Bernoulli Case, Complete Derivation

Target: Prove the Bahadur-Rao (1960) theorem specialized to i.i.d. Bernoulli(p) random variables, with explicit constants and error bounds, and extend to the lower tail and the non-i.i.d. setting.

A.1 Problem Setup

Let $X_1, \dots, X_M \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ with $p \in (0,1)$. Define the sample mean

$$\bar{X}_M = \frac{1}{M} \sum_{i=1}^{M} X_i.$$

For a threshold $\theta > p$, we are interested in the large deviation probability

$$P_M(\theta) := \mathbb{P}(\bar{X}_M \geq \theta).$$

A.2 Cumulant Generating Function and Cramer Transform

The moment generating function of a single Bernoulli(p) variable X is

$$M(\lambda) = \mathbb{E}[e^{\lambda X}] = 1 - p + pe^{\lambda}, \quad \lambda \in \mathbb{R},$$

so the cumulant generating function (CGF) is

$$\psi(\lambda) = \log M(\lambda) = \log(1 - p + pe^{\lambda}).$$

The first two derivatives of ψ are:

$$\psi'(\lambda) = \frac{pe^{\lambda}}{1 - p + pe^{\lambda}},$$

$$\psi''(\lambda) = \frac{pe^{\lambda}(1 - p + pe^{\lambda}) - (pe^{\lambda})^2}{(1 - p + pe^{\lambda})^2} = \frac{p(1-p)e^{\lambda}}{(1 - p + pe^{\lambda})^2}.$$

The rate function (Cramer transform) is the Legendre-Fenchel conjugate of ψ:

$$I(\theta) = \sup_{\lambda \in \mathbb{R}} \{\lambda\theta - \psi(\lambda)\}, \qquad \theta \in [0, 1].$$

For $\theta \in (0, 1)$, the supremum is attained at the unique λ^* solving $\psi'(\lambda) = \theta$:

$$\frac{pe^{\lambda^*}}{1 - p + pe^{\lambda^*}} = \theta.$$

Proposition A.1 (Explicit saddlepoint). *The equation $\psi'(\lambda^*) = \theta$ has the unique solution*

$$\lambda^* = \log \frac{\theta(1-p)}{p(1-\theta)}.$$

Proof. From $\psi'(\lambda^*) = \theta$:

$$\frac{pe^{\lambda^*}}{1-p+pe^{\lambda^*}} = \theta \implies pe^{\lambda^*} = \theta(1-p+pe^{\lambda^*}) \implies pe^{\lambda^*}(1-\theta) = \theta(1-p) \implies e^{\lambda^*} = \frac{\theta(1-p)}{p(1-\theta)}.$$

Taking logs gives the result. $\square$

Proposition A.2 (Rate function = KL divergence). *For* $\theta \in [0,1]$,

$$I(\theta) = \theta \log \frac{\theta}{p} + (1-\theta) \log \frac{1-\theta}{1-p} = \mathrm{KL}(\theta \| p).$$

Proof. Substituting λ^* into $\lambda\theta - \psi(\lambda)$:

$$\begin{aligned}
I(\theta) &= \lambda^*\theta - \log(1-p+pe^{\lambda^*}) \\
&= \theta \log \frac{\theta(1-p)}{p(1-\theta)} - \log\left(1-p+p\cdot\frac{\theta(1-p)}{p(1-\theta)}\right) \\
&= \theta \log \frac{\theta(1-p)}{p(1-\theta)} - \log\left(\frac{(1-p)(1-\theta)+\theta(1-p)}{1-\theta}\right) \\
&= \theta \log \frac{\theta(1-p)}{p(1-\theta)} - \log\left(\frac{1-p}{1-\theta}\right) \\
&= \theta \log \frac{\theta}{p} + (1-\theta) \log \frac{1-\theta}{1-p} = \mathrm{KL}(\theta \| p). \qquad \square
\end{aligned}$$

A.3 Exponential Tilting and the Tilted Distribution

Define the exponentially tilted (Esscher-transformed) probability measure $\mathbb{P}_{\lambda^*}$ by

$$\frac{d\mathbb{P}_{\lambda^*}}{d\mathbb{P}} = \exp(\lambda^* S_M - M\psi(\lambda^*)),$$

where $S_M = \sum_{i=1}^M X_i$. Under $\mathbb{P}_{\lambda^*}$, the X_i are i.i.d. with tilted marginal

$$\mathbb{P}_{\lambda^*}(X_i = x) = \frac{e^{\lambda^* x}}{1 - p + pe^{\lambda^*}} \cdot \mathbb{P}(X_i = x), \qquad x \in \{0, 1\}.$$

Proposition A.3 (Tilted distribution is Bernoulli(θ)). *Under* $\mathbb{P}_{\lambda^*}$,

$$X_i \sim \text{Bernoulli}(\theta), \quad \text{with } \mathbb{E}_{\lambda^*}[X_i] = \theta,\ \text{Var}_{\lambda^*}(X_i) = \sigma^2(\theta) = \theta(1-\theta).$$

Proof. Using $e^{\lambda^*} = \theta(1-p)/[p(1-\theta)]$:

$$\mathbb{P}_{\lambda^*}(X_i = 1) = \frac{e^{\lambda^*} p}{1 - p + pe^{\lambda^*}} = \frac{\frac{\theta(1-p)}{p(1-\theta)} \cdot p}{\frac{1-p}{1-\theta}} = \frac{\theta}{1-\theta} \cdot (1-\theta) = \theta.$$

Thus $\mathbb{P}_{\lambda^*}(X_i = 1) = \theta$, so $X_i \sim \text{Bern}(\theta)$. The variance is $\theta(1-\theta)$.

Alternatively, from the CGF:

$$\psi''(\lambda^*) = \frac{p(1-p)e^{\lambda^*}}{(1-p+pe^{\lambda^*})^2}.$$

Substituting e^{λ^*} and $1 - p + pe^{\lambda^*} = (1-p)/(1-\theta)$:

$$\psi''(\lambda^*) = \frac{p(1-p) \cdot \frac{\theta(1-p)}{p(1-\theta)}}{\left(\frac{1-p}{1-\theta}\right)^2} = \frac{\theta(1-\theta)(1-p)^2/(1-\theta)^2}{(1-p)^2/(1-\theta)^2} = \theta(1-\theta). \qquad \square$$

A.4 The Bahadur-Rao Theorem for Bernoulli

A.4.1 Statement

Theorem A.4 (Bahadur-Rao, 1960; Bernoulli case). *Let* $X_1, \dots, X_M \overset{i.i.d.}{\sim}$ *Bernoulli*(p) *and fix* $\theta > p$. *Then*

$$\mathbb{P}(\bar{X}_M \geq \theta) = \frac{\exp(-M \cdot \mathrm{KL}(\theta \| p))}{\lambda^* \sqrt{2\pi M \cdot \theta(1-\theta)}} \Big(1 + \varepsilon_M\Big),$$

where $\lambda^* = \log \frac{\theta(1-p)}{p(1-\theta)} > 0$ *and*

$$|\varepsilon_M| \leq \frac{C(p,\theta)}{M}$$

for an explicit constant $C(p,\theta)$ *given in Proposition A.5 below.*

Remark. The result also holds for the strict inequality $\bar{X}_M > \theta$, since $\mathbb{P}(\bar{X}_M = \theta) = 0$ for non-integer $M\theta$, and for integer $M\theta$ the correction is absorbed into the $O(1/M)$ term.

A.4.2 Proof via Stirling's Formula (Direct Binomial Tail Expansion)

Since $S_M = \sum_{i=1}^{M} X_i \sim \text{Binomial}(M, p)$, we can work directly with the binomial distribution. Let $k = \lceil M\theta \rceil$. Since $\theta > p$ and θ is fixed, we have $k > Mp$ for all sufficiently large M.

The binomial tail probability is

$$P_M(\theta) = \mathbb{P}(S_M \geq k) = \sum_{j=k}^{M} a_j, \qquad a_j = \binom{M}{j} p^j (1-p)^{M-j}.$$

Step 1: Stirling's formula with explicit remainder.

For any $n \in \mathbb{N}$, Robbins' refinement of Stirling gives

$$\sqrt{2\pi n}\left(\frac{n}{e}\right)^n \exp\!\left(\frac{1}{12n+1}\right) < n! < \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \exp\!\left(\frac{1}{12n}\right).$$

Thus $n! = \sqrt{2\pi n}\,(n/e)^n\, e^{r_n}$ with $|r_n| \leq 1/(12n)$ for $n \geq 1$.

Applying this to $\binom{M}{k} = M!/(k!\,(M-k)!)$:

$$\binom{M}{k} = \sqrt{\frac{M}{2\pi k(M-k)}}\,\frac{M^M}{k^k(M-k)^{M-k}}\,\exp(r_M - r_k - r_{M-k}).$$

The error term satisfies

$$|r_M - r_k - r_{M-k}| \leq \frac{1}{12M} + \frac{1}{12k} + \frac{1}{12(M-k)} \leq \frac{1}{4\min(k, M-k)}.$$

Define $\theta_M = k/M$. Writing $H(x) = -x\log x - (1-x)\log(1-x)$ (binary entropy):

$$\binom{M}{k} = \frac{\exp\big(M \cdot H(\theta_M)\big)}{\sqrt{2\pi M \cdot \theta_M(1-\theta_M)}} \cdot (1 + \delta_M^{(1)}), \qquad |\delta_M^{(1)}| \leq \frac{C_1}{M}$$

for an explicit constant $C_1 = C_1(p,\theta)$ that depends on p and θ but not on M. (Here we bound $|e^{r_M - r_k - r_{M-k}} - 1|$ using $|e^u - 1| \leq |u|e^{|u|}$.)

Step 2: Leading term a_k.

$$\begin{aligned} a_k &= \binom{M}{k} p^k (1-p)^{M-k} \\ &= \frac{\exp\big(-M \cdot I(\theta_M)\big)}{\sqrt{2\pi M \cdot \theta_M (1-\theta_M)}} \cdot (1+\delta_M^{(1)}), \end{aligned}$$

where $I(x) = x \log \frac{x}{p} + (1-x) \log \frac{1-x}{1-p} = \mathrm{KL}(x \| p)$. Since $|\theta_M - \theta| \le 1/M$, a Taylor expansion gives

$$I(\theta_M) = I(\theta) + I'(\theta)(\theta_M - \theta) + O(M^{-2}),$$

with $I'(\theta) = \log \frac{\theta(1-p)}{p(1-\theta)} = \lambda^*$. Hence

$$e^{-M \cdot I(\theta_M)} = e^{-M \cdot I(\theta)} \cdot \exp\big(-M\lambda^*(\theta_M - \theta) + O(1/M)\big).$$

Since $\theta_M - \theta = O(1/M)$, the term $-M\lambda^*(\theta_M - \theta) = O(1)$. Combining all $O(1/M)$ errors:

$$a_k = \frac{\exp\big(-M \cdot I(\theta)\big)}{\sqrt{2\pi M \cdot \theta(1-\theta)}} \cdot (1+\varepsilon_M^{(1)}), \qquad |\varepsilon_M^{(1)}| \le \frac{C_2}{M}. \tag{A.1}$$

Step 3: Ratio of consecutive terms and geometric decay.

For $j \ge k$,

$$\frac{a_{j+1}}{a_j} = \frac{M-j}{j+1} \cdot \frac{p}{1-p}.$$

For the term at $j = k$:

$$\frac{a_{k+1}}{a_k} = \frac{M-k}{k+1} \cdot \frac{p}{1-p} = e^{-\lambda_k^*} \cdot \frac{k}{k+1}, \qquad \lambda_k^* = \log \frac{k(1-p)}{p(M-k)}.$$

Since $k = M\theta_M$, we have $\lambda_k^* = \lambda^* + O(1/M)$, and $k/(k+1) = 1 - 1/k = 1 + O(1/M)$. Thus

$$\frac{a_{k+1}}{a_k} = e^{-\lambda^*}(1 + O(1/M)).$$

Moreover, for all $j \geq k$, the ratio is decreasing in j (since $(M-j)/(j+1)$ decreases with j), so

$$\frac{a_{j+1}}{a_j} \leq \frac{a_{k+1}}{a_k} \leq e^{-\lambda^*}(1 + \rho_M),$$

where $\rho_M \to 0$ as $M \to \infty$ and $|\rho_M| \leq C_3/M$ for an explicit C_3.

Step 4: Summing the tail.

We bound $P_M(\theta) = a_k + a_{k+1} + \cdots + a_M$ from both sides.

Upper bound:

$$P_M(\theta) \leq a_k \sum_{j=0}^{\infty} \left(e^{-\lambda^*}(1 + \rho_M)\right)^j = \frac{a_k}{1 - e^{-\lambda^*}(1 + \rho_M)}.$$

Lower bound:

$$P_M(\theta) \geq a_k \sum_{j=0}^{J} \left(e^{-\lambda^*}(1 - \tilde{\rho}_M)\right)^j$$

for any finite J (where $\tilde{\rho}_M$ accounts for the lower bound on the ratio). Taking $J \to \infty$ (valid since the series converges), we have

$$P_M(\theta) \geq \frac{a_k}{1 - e^{-\lambda^*}(1 - \tilde{\rho}_M)}.$$

Combining both bounds and using $|\rho_M|, |\tilde{\rho}_M| \leq C_3/M$:

$$P_M(\theta) = \frac{a_k}{1 - e^{-\lambda^*}} \cdot (1 + \varepsilon_M^{(2)}), \qquad |\varepsilon_M^{(2)}| \leq \frac{C_4}{M}. \tag{A.2}$$

Step 5: Final assembly.

Substituting (A.1) into (A.2):

$$P_M(\theta) = \frac{\exp(-M \cdot I(\theta))}{(1 - e^{-\lambda^*})\sqrt{2\pi M \cdot \theta(1-\theta)}} \cdot (1 + \varepsilon_M), \qquad |\varepsilon_M| \leq \frac{C(p,\theta)}{M}.$$

Lattice correction factor. The factor $(1 - e^{-\lambda^*})^{-1}$ is the **lattice correction** characteristic of lattice distributions (span $h = 1$). For large λ^* (large deviations), $e^{-\lambda^*}$ is small and $(1 - e^{-\lambda^*})^{-1} \approx 1 + e^{-\lambda^*}$ differs negligibly from 1. For moderate λ^*, the correction matters for exact constants.

Relation to the $1/\lambda^*$ form. Since $1 - e^{-\lambda^*} = \lambda^* - \lambda^{*2}/2 + \lambda^{*3}/6 - \cdots$,

$$\frac{1}{1 - e^{-\lambda^*}} = \frac{1}{\lambda^*}\left(1 + \frac{\lambda^*}{2} + \frac{\lambda^{*2}}{12} - \frac{\lambda^{*4}}{720} + \cdots\right).$$

Both forms are asymptotically equivalent as $\lambda^* \to 0$ (i.e., as $\theta \downarrow p$), but differ by a constant factor for fixed $\lambda^* > 0$. For consistency with the proof architecture (Section 2.4 of the parent document), we write

$$\boxed{\mathbb{P}(\bar{X}_M \geq \theta) = \frac{\exp(-M \cdot \mathrm{KL}(\theta \| p))}{\lambda^* \sqrt{2\pi M \cdot \theta(1-\theta)}} \cdot \frac{\lambda^*}{1-e^{-\lambda^*}} \cdot (1 + O(1/M)).}$$

The factor $\lambda^*/(1-e^{-\lambda^*})$ is a known constant in $(0, \infty)$ and is absorbed into the leading constant in the F1 expansion (where it appears in both numerator and denominator of ratios, so its effect can be tracked explicitly if needed). □

Remark (Connection to the Bahadur-Rao theorem). The general Bahadur-Rao (1960) theorem for non-lattice distributions yields the $1/\lambda^*$ form directly; the lattice correction $(1-e^{-\lambda^*})^{-1}$ appears for lattice distributions like Bernoulli. Both forms agree in the $O(1/M)$ asymptotic expansion up to a constant factor, and the $O(1/M)$ error bound above holds in either case.

A.4.3 Explicit Constant for the $O(1/M)$ Bound

Proposition A.5 (Explicit $O(1/M)$ bound for Bernoulli). For $p \in (0,1)$, $\theta > p$, let $\lambda^* = \log \frac{\theta(1-p)}{p(1-\theta)}$, $\sigma^2 = \theta(1-\theta)$, and $k = \lceil M\theta \rceil$. Then for M large enough that $\min(k, M-k) \geq 1$,

$$\left| \frac{\mathbb{P}(\bar{X}_M \geq \theta) \cdot (1-e^{-\lambda^*})\sqrt{2\pi M}\,\sigma}{e^{-M \cdot \mathrm{KL}(\theta \| p)}} - 1 \right| \leq \frac{C_{\mathrm{Stirling}}(p, \theta)}{M},$$

where

$$C_{\mathrm{Stirling}}(p, \theta) = \frac{1}{4\min(\theta, 1-\theta)} + \frac{C_1(p, \theta)}{\sqrt{2\pi}} + \frac{C_3(p, \theta)}{(1-e^{-\lambda^*})^2},$$

and C_1, C_3 are the explicit constants from Steps 2 and 4 of the proof above, bounded in terms of p and θ only.

Proof. The Stirling remainder contributes $|\delta_M^{(1)}| \leq \frac{1}{4\min(k,M-k)}$, the $k/M \to \theta$ replacement contributes $O(1/M)$ by Lipschitz continuity of I and the square-root factor, and the geometric-series approximation contributes $|\varepsilon_M^{(2)}| \leq C_3/(M(1-e^{-\lambda^*}))$. Collecting these gives the bound. □

This constant is fully explicit: for any given p and θ, all quantities can be evaluated numerically.

A.5 Lower Tail Formula ($\theta < p$)

For the **lower tail** $\mathbb{P}(\bar{X}_M \leq \theta)$ with $\theta < p$, we use the symmetry transformation $Y_i = 1 - X_i \sim \text{Bern}(1-p)$. Then

$$\bar{X}_M \leq \theta \iff \frac{1}{M}\sum_{i=1}^{M} Y_i \geq 1 - \theta,$$

and $1 - \theta > 1 - p$ since $\theta < p$. Applying Theorem A.4 to the Y_i sequence gives the following.

Theorem A.6 (Lower tail Bahadur-Rao for Bernoulli). Let $X_1, \ldots, X_M \overset{i.i.d.}{\sim} \text{Bernoulli}(p)$ and fix $\theta < p$. Then

$$\mathbb{P}(\bar{X}_M \leq \theta) = \frac{\exp(-M \cdot \text{KL}(\theta \| p))}{|\lambda^*|\sqrt{2\pi M \cdot \theta(1-\theta)}}\Big(1 + \varepsilon'_M\Big),$$

where

$$\lambda^* = \log \frac{\theta(1-p)}{p(1-\theta)} < 0, \qquad |\lambda^*| = \log \frac{p(1-\theta)}{\theta(1-p)},$$

and $|\varepsilon'_M| \leq C'(p,\theta)/M$ for an explicit constant C'.

Proof. Apply Theorem A.4 to $Y_i = 1 - X_i \sim \text{Bern}(1-p)$ with threshold $1-\theta > 1-p$. The saddlepoint for the Y-problem is

$$\lambda_Y^* = \log \frac{(1-\theta)(1-(1-p))}{(1-p)(1-(1-\theta))} = \log \frac{(1-\theta)p}{(1-p)\theta} = -\lambda^* > 0.$$

The rate function:

$$\text{KL}(1-\theta \| 1-p) = (1-\theta)\log\frac{1-\theta}{1-p} + \theta \log \frac{\theta}{p} = \text{KL}(\theta \| p).$$

The variance: $(1-\theta)\theta = \theta(1-\theta)$, unchanged. Substituting into Theorem A.4 gives the result. □

Application to FNR. In the noise-detection problem where $e_m \sim \text{Bern}(p_1)$ under H_1 and $p_1 > \theta$:

$$\text{FNR}_M = \mathbb{P}(C_M \leq \theta \mid \text{H}_1) \sim \frac{\exp(-M \cdot \text{KL}(\theta \| p_1))}{|\lambda_1^*|\sqrt{2\pi M \cdot \theta(1-\theta)}},$$

where $\lambda_1^* = \log \frac{\theta(1-p_1)}{p_1(1-\theta)} < 0$, $|\lambda_1^*| = \log \frac{p_1(1-\theta)}{\theta(1-p_1)}$.

A.6 Non-i.i.d. Extension

A.6.1 Setting

Consider independent but **not identically distributed** Bernoulli variables:

$$X_m \sim \text{Bernoulli}(p_m), \qquad m = 1, \dots, M,$$

with $p_m \in [p_{\min}, p_{\max}] \subset (0, 1)$ bounded away from 0 and 1. Define the weighted sample mean $C_M = \frac{1}{M}\sum_{m=1}^{M} X_m$ and a threshold $\theta > \limsup_{M\to\infty} \frac{1}{M}\sum p_m$.

A.6.2 What Changes

1. **Cumulant generating function** (now depends on M):

$$\psi_M(\lambda) = \frac{1}{M}\sum_{m=1}^{M} \log(1 - p_m + p_m e^{\lambda}).$$

2. **Saddlepoint** λ_M^* solves $\psi_M'(\lambda) = \theta$:

$$\frac{1}{M}\sum_{m=1}^{M} \frac{p_m e^{\lambda_M^*}}{1 - p_m + p_m e^{\lambda_M^*}} = \theta.$$

   This is no longer a closed form, but λ_M^* is bounded and strictly positive.

3. **Rate function** (Gartner-Ellis):

$$I_M(\theta) = \sup_{\lambda}\{\lambda\theta - \psi_M(\lambda)\} = \lambda_M^*\theta - \psi_M(\lambda_M^*).$$

This is **not** equal to $\mathrm{KL}(\theta \| p)$ for any single p.

4. **Tilted distribution**: Under $\mathbb{P}_{\lambda_M^*}$, the X_m remain independent with

$$\mathbb{P}_{\lambda_M^*}(X_m = 1) = \frac{p_m e^{\lambda_M^*}}{1 - p_m + p_m e^{\lambda_M^*}} =: q_m(\theta).$$

The tilted mean is $\frac{1}{M} \sum q_m(\theta) = \theta$ by construction. The tilted variance is

$$\sigma_M^2(\theta) = \psi_M''(\lambda_M^*) = \frac{1}{M} \sum_{m=1}^{M} \frac{p_m(1-p_m)e^{\lambda_M^*}}{(1 - p_m + p_m e^{\lambda_M^*})^2} = \frac{1}{M} \sum_{m=1}^{M} q_m(\theta)(1-q_m(\theta)).$$

Note: $q_m(\theta) \in (p'_{\min}, p'_{\max})$ bounded away from 0,1.

5. **Bahadur-Rao analogue** (heterogeneous case): Under the Lindeberg-Feller CLT for the tilted measure,

$$\mathbb{P}(C_M \geq \theta) = \frac{\exp(-M \cdot I_M(\theta))}{\lambda_M^* \sqrt{2\pi M \cdot \sigma_M^2(\theta)}} \Big(1 + O(1/M)\Big),$$

provided the p_m sequence is such that the $q_m(\theta)$ satisfy a Lyapunov or Lindeberg condition (which holds automatically when $p_m \in [p_{\min}, p_{\max}]$).

The $O(1/M)$ constant depends on $\max_m p_m(1-p_m)$ and $\min_m p_m(1-p_m)$.

A.6.3 Summary of Changes

Quantity	i.i.d. case	Non-i.i.d. case
Saddlepoint λ^*	Closed form $\log \frac{\theta(1-p)}{p(1-\theta)}$	Solves $\frac{1}{M}\sum \frac{p_m e^{\lambda}}{1-p_m+p_m e^{\lambda}} = \theta$
Rate function $I(\theta)$	$\mathrm{KL}(\theta \\| p)$	$\lambda_M^* \theta - \frac{1}{M}\sum \log(1-p_m + p_m e^{\lambda_M^*})$
Tilted variance $\sigma^2(\theta)$	$\theta(1-\theta)$	$\frac{1}{M}\sum q_m(\theta)(1-q_m(\theta))$
Leading constant denominator	λ^*	λ_M^*

The **functional form** of the Bahadur-Rao expansion remains identical; only the constants (λ^*, σ^2, I) are replaced by their M-dependent counterparts.

A.7 Summary for Lemma A

Quantity	Formula for i.i.d. Bern(p), $\theta > p$
CGF $\psi(\lambda)$	$\log(1-p+pe^{\lambda})$
Saddlepoint λ^*	$\log \frac{\theta(1-p)}{p(1-\theta)} > 0$
Rate function $I(\theta)$	$\mathrm{KL}(\theta \\| p) = \theta \log \frac{\theta}{p} + (1-\theta)\log \frac{1-\theta}{1-p}$
Tilted mean	θ
Tilted variance $\sigma^2(\theta)$	$\theta(1-\theta)$

Quantity	Formula for i.i.d. Bern(p), $\theta > p$
Upper tail ($\theta > p$)	$\frac{e^{-M\cdot\mathrm{KL}(\theta\|p)}}{\lambda^*\sqrt{2\pi M\theta(1-\theta)}}(1+O(1/M))$
Lower tail ($\theta < p$)	$\frac{e^{-M\cdot\mathrm{KL}(\theta\|p)}}{\lvert\lambda^*\rvert\sqrt{2\pi M\theta(1-\theta)}}(1+O(1/M))$
$O(1/M)$ constant	Explicit in terms of p, θ (Proposition A.5)

Lemma B: F1 Asymptotic Expansion with Higher-Order Terms

Target: Starting from the SCX F1 formula, derive the exact asymptotic expansion of $1-\mathrm{F1}$ in terms of FPR and FNR, verify the expansion's validity at the boundaries $\eta \to 0$ and $\eta \to 1/2$, and substitute the Bahadur-Rao asymptotics to obtain the leading constant.

B.1 Setup and Notation

Recall the SCX noise-detection setup:

- $\eta = \mathbb{P}(\text{noise})$: prior probability of a noisy sample.
- $\mathrm{TPR} = 1 - \mathrm{FNR}_M$: true positive rate (detection power).

- $\mathrm{FPR} = \mathrm{FPR}_M$: false positive rate (type I error).
- $C_M = \frac{1}{M}\sum e_m$: consensus score.

The F1 score is defined as the harmonic mean of precision and recall:

$$\mathrm{F1} = \frac{2\eta \cdot \mathrm{TPR}}{\eta(1+\mathrm{TPR}) + (1-\eta)\mathrm{FPR}}.$$

Verification of the formula. Using the standard definitions:

$$\text{Precision} = \frac{\eta \cdot \mathrm{TPR}}{\eta \cdot \mathrm{TPR} + (1-\eta)\mathrm{FPR}}, \qquad \text{Recall} = \mathrm{TPR},$$

we compute

$$\begin{aligned}
\mathrm{F1} &= \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \\
&= \frac{2 \cdot \frac{\eta \cdot \mathrm{TPR}}{\eta \cdot \mathrm{TPR} + (1-\eta)\mathrm{FPR}} \cdot \mathrm{TPR}}{\frac{\eta \cdot \mathrm{TPR}}{\eta \cdot \mathrm{TPR} + (1-\eta)\mathrm{FPR}} + \mathrm{TPR}} \\
&= \frac{2\eta \cdot \mathrm{TPR}^2}{\eta \cdot \mathrm{TPR} + (1-\eta)\mathrm{FPR}} \cdot \frac{1}{\frac{\eta \cdot \mathrm{TPR}}{\eta \cdot \mathrm{TPR} + (1-\eta)\mathrm{FPR}} + \mathrm{TPR}} \\
&= \frac{2\eta \cdot \mathrm{TPR}^2}{\eta \cdot \mathrm{TPR} + \mathrm{TPR}(\eta \cdot \mathrm{TPR} + (1-\eta)\mathrm{FPR})} \\
&= \frac{2\eta \cdot \mathrm{TPR}}{\eta + \eta \cdot \mathrm{TPR} + (1-\eta)\mathrm{FPR}} \\
&= \frac{2\eta \cdot \mathrm{TPR}}{\eta(1+\mathrm{TPR}) + (1-\eta)\mathrm{FPR}}.
\end{aligned}$$

This matches the expression in the proof architecture. No algebraic error is present – the denominator is always positive for $\eta > 0$.

B.2 Exact Expression for $1 - \mathbf{F1}$

Substituting $\mathrm{TPR} = 1 - \mathrm{FNR}$:

$$\mathrm{F1} = \frac{2\eta(1-\mathrm{FNR})}{\eta(2-\mathrm{FNR}) + (1-\eta)\mathrm{FPR}}.$$

Thus

$$\begin{aligned}
1 - \mathrm{F1} &= 1 - \frac{2\eta(1-\mathrm{FNR})}{\eta(2-\mathrm{FNR}) + (1-\eta)\mathrm{FPR}} \\
&= \frac{\eta(2-\mathrm{FNR}) + (1-\eta)\mathrm{FPR} - 2\eta(1-\mathrm{FNR})}{\eta(2-\mathrm{FNR}) + (1-\eta)\mathrm{FPR}} \\
&= \frac{\eta \cdot \mathrm{FNR} + (1-\eta)\mathrm{FPR}}{\eta(2-\mathrm{FNR}) + (1-\eta)\mathrm{FPR}}.
\end{aligned}$$

Sanity checks: - $\mathrm{FNR} = \mathrm{FPR} = 0 \implies 1 - \mathrm{F1} = 0$. - $\eta = 0$: $1 - \mathrm{F1} = \mathrm{FPR}/\mathrm{FPR} = 1$ ($\mathrm{F1} = 0$, no positives possible). - $\eta = 1$: $1 - \mathrm{F1} = \mathrm{FNR}/(2 - \mathrm{FNR}) \approx \mathrm{FNR}/2$. - Denominator $= 2\eta - \eta \cdot \mathrm{FNR} + (1-\eta)\mathrm{FPR} \geq \eta > 0$ (strictly positive for $\eta > 0$).

B.3 First-Order Expansion

Let

$$\alpha = \eta \cdot \mathrm{FNR} + (1-\eta)\mathrm{FPR}, \qquad \beta = 2\eta - \eta \cdot \mathrm{FNR} + (1-\eta)\mathrm{FPR}.$$

Then

$$1 - \mathrm{F1} = \frac{\alpha}{\beta}.$$

Factor $\beta = 2\eta(1-\gamma)$ where

$$\gamma := \frac{\eta \cdot \mathrm{FNR} - (1-\eta)\mathrm{FPR}}{2\eta} = \frac{\mathrm{FNR}}{2} - \frac{(1-\eta)\mathrm{FPR}}{2\eta}.$$

Thus

$$1 - \mathrm{F1} = \frac{\alpha}{2\eta} \cdot \frac{1}{1-\gamma}.$$

For $|\gamma| < 1$ (which holds for all sufficiently large M since $\mathrm{FNR}, \mathrm{FPR} \to 0$), the geometric series converges:

$$\frac{1}{1-\gamma} = \sum_{k=0}^{\infty} \gamma^k.$$

Let

$$A := \frac{\mathrm{FNR}}{2}, \qquad B := \frac{(1-\eta)\mathrm{FPR}}{2\eta}.$$

Then $\alpha/(2\eta) = A + B$, and $\gamma = A - B$.

First-order expansion:

$$1 - \mathrm{F1} = A + B + R,$$

where

$$R = (A + B) \cdot \frac{\gamma}{1-\gamma} = \frac{(A+B)(A-B)}{1-(A-B)}.$$

Theorem B.1 (First-order expansion with error bound). For $|\gamma| = |A - B| < 1$,

$$1 - \mathrm{F1} = \frac{\mathrm{FNR}}{2} + \frac{(1-\eta)}{2\eta}\,\mathrm{FPR} + R,$$

where the remainder satisfies

$$|R| \le \frac{(\mathrm{FNR}/2 + (1-\eta)\mathrm{FPR}/(2\eta))^2}{1 - \mathrm{FNR}/2 - (1-\eta)\mathrm{FPR}/(2\eta)}.$$

Proof. We have $|A + B| \le r$ and $|A - B| \le r$ where $r = A + B = \mathrm{FNR}/2 + (1-\eta)\mathrm{FPR}/(2\eta)$. Then

$$|R| = \frac{(A+B)|A-B|}{1-|A-B|} \le \frac{r^2}{1-r}.$$

For sufficiently large M, $r \le 1/2$ and $|R| \le 2r^2$. This gives the bound. □

Corollary B.2 (Quadratic bound). For M large enough that $r \le 1/2$,

$$|R| \le 2\left(\frac{\mathrm{FNR}}{2} + \frac{(1-\eta)\mathrm{FPR}}{2\eta}\right)^2 \le \frac{\mathrm{FNR}^2}{2} + \frac{(1-\eta)\mathrm{FNR}\cdot\mathrm{FPR}}{\eta} + \frac{(1-\eta)^2\mathrm{FPR}^2}{2\eta^2}.$$

In particular, $R = O(\max(\mathrm{FPR}^2, \mathrm{FNR}^2, \mathrm{FPR}\cdot\mathrm{FNR}))$.

B.4 Second-Order Expansion (Explicit Terms)

Expanding to second order in γ:

$$\begin{aligned} 1 - \mathrm{F1} &= (A+B)(1+\gamma+\gamma^2+\gamma^3+\cdots) \\ &= (A+B) + (A+B)(A-B) + (A+B)(A-B)^2 + O(r^4). \end{aligned}$$

Explicit second-order terms:

$$\begin{aligned} 1 - \mathrm{F1} = \frac{\mathrm{FNR}}{2} &+ \frac{(1-\eta)\mathrm{FPR}}{2\eta} \\ &+ \frac{\mathrm{FNR}^2}{4} - \frac{(1-\eta)^2\mathrm{FPR}^2}{4\eta^2} \\ &+ \left(\tfrac{\mathrm{FNR}}{2} + \tfrac{(1-\eta)\mathrm{FPR}}{2\eta}\right)\left(\tfrac{\mathrm{FNR}}{2} - \tfrac{(1-\eta)\mathrm{FPR}}{2\eta}\right)^2 \\ &+ O(r^4). \end{aligned}$$

Explicit $O(\cdot)$ constant in terms of $\max(\mathbf{FPR}^2, \mathbf{FNR}^2, \mathbf{FPR}\cdot\mathbf{FNR})$:

From Corollary B.2, when $r \leq 1/2$:

$$|R| \leq \frac{\mathrm{FNR}^2}{2} + \frac{(1-\eta)\mathrm{FNR}\cdot\mathrm{FPR}}{\eta} + \frac{(1-\eta)^2\mathrm{FPR}^2}{2\eta^2}.$$

Let $M_2 = \max(\mathrm{FNR}^2, \mathrm{FPR}^2, \mathrm{FNR}\cdot\mathrm{FPR})$. Then

$$|R| \leq \left(\frac{1}{2} + \frac{1-\eta}{\eta} + \frac{(1-\eta)^2}{2\eta^2}\right) M_2 = \frac{1}{2\eta^2}(\eta^2 + 2\eta(1-\eta) + (1-\eta)^2) M_2 = \frac{1}{2\eta^2}\cdot 1 \cdot M_2,$$

where we used $\eta^2 + 2\eta(1-\eta) + (1-\eta)^2 = (\eta + (1-\eta))^2 = 1$.

Thus

$$|R| \le \frac{1}{2\eta^2} \cdot \max(\mathrm{FNR}^2, \mathrm{FPR}^2, \mathrm{FNR} \cdot \mathrm{FPR}).$$

This gives the explicit constant $C = 1/(2\eta^2)$ for the $O(\cdot)$ remainder.

B.5 Substituting the Bahadur-Rao Asymptotics

Now substitute the leading-order Bahadur-Rao expansions:

$$\mathrm{FPR}_M \sim \frac{e^{-M\kappa_0}}{\lambda_0^* \sqrt{2\pi M\theta(1-\theta)}}, \quad \kappa_0 = \mathrm{KL}(\theta \| p_0),\ \lambda_0^* = \log\frac{\theta(1-p_0)}{p_0(1-\theta)} > 0,$$
$$\mathrm{FNR}_M \sim \frac{e^{-M\kappa_1}}{|\lambda_1^*| \sqrt{2\pi M\theta(1-\theta)}}, \quad \kappa_1 = \mathrm{KL}(\theta \| p_1),\ |\lambda_1^*| = \log\frac{p_1(1-\theta)}{\theta(1-p_1)} > 0.$$

Theorem B.3 (Leading term of $1 - \mathbf{F1}$ with Bahadur-Rao).

$$1\text{–F1} = \frac{1}{\sqrt{2\pi M \cdot \theta(1-\theta)}} \left[\frac{e^{-M\kappa_1}}{2\,|\lambda_1^*|} + \frac{1-\eta}{2\eta} \cdot \frac{e^{-M\kappa_0}}{\lambda_0^*} \right] + O\left(\frac{e^{-2M\min(\kappa_0,\kappa_1)}}{M} \right).$$

Proof. Substitute the leading terms of FPR and FNR into the first-order expansion from Theorem B.1. The remainder R from Corollary B.2 is bounded by $O(\max(\mathrm{FPR}^2, \mathrm{FNR}^2)) = O(e^{-2M\min(\kappa_0,\kappa_1)}/M)$, since each of FPR^2 and FNR^2 decays with twice the exponent and an extra $1/M$ factor from the square. □

B.5.1 Optimal Threshold: Chernoff Point

The optimal threshold θ^* balances the two exponents:

$$\kappa = \mathrm{KL}(\theta^* \| p_0) = \mathrm{KL}(\theta^* \| p_1) = C(\mathrm{Bern}(p_0), \mathrm{Bern}(p_1)),$$

where $C(\cdot, \cdot)$ is the Chernoff information. At $\theta = \theta^*$:

$$1 - \mathrm{F1} \sim \frac{e^{-M\kappa}}{\sqrt{2\pi M \cdot \theta^*(1-\theta^*)}} \cdot \left[\frac{1}{2|\lambda_1^*(\theta^*)|} + \frac{1-\eta}{2\eta \cdot \lambda_0^*(\theta^*)} \right].$$

B.6 Verification of the Expansion at Special Cases

B.6.1 Case $\eta \to 0$ (Sparse Noise)

As $\eta \to 0$, the coefficient $(1-\eta)/(2\eta) \to \infty$, so the FPR term dominates both the FNR term and the expansion itself. The convergence condition $|\gamma| < 1$ becomes:

$$|\gamma| = \left| \frac{\mathrm{FNR}}{2} - \frac{(1-\eta)\mathrm{FPR}}{2\eta} \right| < 1.$$

For small η, this is dominated by the second term, requiring $\mathrm{FPR} < 2\eta/(1-\eta) \approx 2\eta$.

Using the Bahadur-Rao approximation $\mathrm{FPR} \approx e^{-M\kappa_0}/(\lambda_0^* \sqrt{M})$, the condition becomes:

$$\frac{e^{-M\kappa_0}}{\lambda_0^* \sqrt{2\pi M\theta(1-\theta)}} \lesssim 2\eta.$$

Definition (Threshold $\eta_{\min}$). Define

$$\eta_{\min}(M) = \frac{1}{2} \cdot \frac{e^{-M\kappa_0}}{\lambda_0^* \sqrt{2\pi M\theta(1-\theta)}}.$$

For $\eta \ll \eta_{\min}(M)$, the geometric series $\sum \gamma^k$ converges slowly (or diverges), and the polynomial expansion in FPR/FNR is no longer valid. In this regime, a different asymptotic expression is needed:

$$\begin{aligned} 1 - \mathrm{F1} &= \frac{\eta \cdot \mathrm{FNR} + (1-\eta)\mathrm{FPR}}{\eta(2 - \mathrm{FNR}) + (1-\eta)\mathrm{FPR}} \\ &\approx \frac{\mathrm{FPR}}{\mathrm{FPR} + \eta/(1-\eta)} \qquad \text{(neglecting FNR terms)} \\ &= \frac{1}{1 + \eta/[(1-\eta)\mathrm{FPR}]}. \end{aligned}$$

When $\eta \ll \eta_{\min}$, we have $\eta \ll \mathrm{FPR}$, so $1-\mathrm{F1} \approx 1$ ($\mathrm{F1} \approx 0$). This is intuitive: with extremely rare noise, even a tiny FPR destroys precision.

Practical implication: The expansion $1-\mathrm{F1} \approx \mathrm{FNR}/2+(1-\eta)\mathrm{FPR}/(2\eta)$ is valid whenever

$$\eta \gtrsim \eta_{\min}(M) = \frac{e^{-M\kappa_0}}{2\lambda_0^* \sqrt{2\pi M\theta(1-\theta)}}.$$

For realistic M, this threshold is exponentially small, so the expansion holds in almost all practical settings.

B.6.2 Case $\eta \to 1/2$ (Balanced Noise)

When $\eta = 1/2$, the expansion simplifies dramatically:

$$1 - \mathrm{F1} = \frac{\frac{1}{2}\mathrm{FNR} + \frac{1}{2}\mathrm{FPR}}{\frac{1}{2}(2 - \mathrm{FNR}) + \frac{1}{2}\mathrm{FPR}} = \frac{\mathrm{FNR} + \mathrm{FPR}}{2 - \mathrm{FNR} + \mathrm{FPR}}$$
$$= \frac{\mathrm{FNR} + \mathrm{FPR}}{2} \cdot \frac{1}{1 - (\mathrm{FNR} - \mathrm{FPR})/2}.$$

First-order:

$$1 - \mathrm{F1} = \frac{\mathrm{FNR} + \mathrm{FPR}}{2} + O(\max(\mathrm{FNR}^2, \mathrm{FPR}^2)).$$

With Bahadur-Rao substitution at θ^*:

$$1 - \mathrm{F1} \sim \frac{e^{-M\kappa}}{2\sqrt{2\pi M \cdot \theta^*(1-\theta^*)}} \left(\frac{1}{|\lambda_1^*|} + \frac{1}{\lambda_0^*} \right).$$

In the symmetric case $p_0 = 1 - p_1$ (which implies $\theta^* = 1/2$, $\lambda_0^* = |\lambda_1^*|$), this further simplifies to:

$$1 - \mathrm{F1} \sim \frac{e^{-M\kappa}}{\lambda_0^* \sqrt{2\pi M \cdot \theta^*(1-\theta^*)}}.$$

B.6.3 Summary of Regimes

Regime	Leading term of $1 - \mathrm{F1}$
$\eta \gg \eta_{\min}$ (general)	$\frac{\mathrm{FNR}}{2} + \frac{1-\eta}{2\eta}\mathrm{FPR}$

Regime	Leading term of $1 - \mathrm{F1}$
$\eta = 1/2$ (balanced)	$(\mathrm{FNR} + \mathrm{FPR})/2$
$\eta \ll \eta_{\min}$ (ultra-sparse)	≈ 1 (expansion breaks down)
$\eta \to 1$ (dominant noise)	$\mathrm{FNR}/2 + o(\mathrm{FNR})$

B.7 Explicit Constant for the $O(\cdot)$ Term

Collecting the results from Sections B.3-B.5, we have the complete asymptotic:

Theorem B.4 (Full expansion with explicit constants). Let $\mathrm{FPR} = \mathrm{FPR}_M$ and $\mathrm{FNR} = \mathrm{FNR}_M$ be the error rates from the threshold test with threshold θ. Assume M is large enough that

$$r := \frac{\mathrm{FNR}}{2} + \frac{(1-\eta)\mathrm{FPR}}{2\eta} \le \frac{1}{2}.$$

Then

$$1 - \mathrm{F1} = \frac{\mathrm{FNR}}{2} + \frac{1-\eta}{2\eta}\mathrm{FPR} + \frac{\mathrm{FNR}^2}{4} - \frac{(1-\eta)^2\mathrm{FPR}^2}{4\eta^2} + R_3,$$

where $|R_3| \le 2r^3 \le \frac{1}{4\eta^3}\big(\eta \cdot \mathrm{FNR} + (1-\eta)\mathrm{FPR}\big)^3$.

When $\eta \gtrsim \eta_{\min}(M)$, substituting the Bahadur-Rao expansions yields:

$$1-\mathrm{F1} = \frac{1}{\sqrt{2\pi M\theta(1-\theta)}}\left[\frac{e^{-M\kappa_1}}{2|\lambda_1^*|} + \frac{1-\eta}{2\eta}\frac{e^{-M\kappa_0}}{\lambda_0^*}\right]\cdot(1+O(1/M))+O\left(\frac{e^{-2M\min(\kappa_0,\kappa_1)}}{M}\right).$$

At the Chernoff point θ^* where $\kappa_0 = \kappa_1 = \kappa$:

$$1 - \mathrm{F1} = \frac{e^{-M\kappa}}{\sqrt{2\pi M\theta^*(1-\theta^*)}}\left[\frac{1}{2|\lambda_1^*|} + \frac{1-\eta}{2\eta\lambda_0^*}\right]\cdot(1+O(1/M)).$$

B.8 Verification: No Denominator Bug

The user noted a concern about the denominator going negative in v1. We verify:

$$\text{Denominator} = \eta(2-\mathrm{FNR}) + (1-\eta)\mathrm{FPR}.$$

Since FNR ≤ 1, we have $2 - \mathrm{FNR} \geq 1$, so

$$\text{Denominator} \geq \eta\cdot 1 + 0 = \eta > 0 \quad (\text{for } \eta > 0).$$

For $\eta = 0$, the F1 formula is degenerate (no positive samples), which is handled separately. **No algebraic error exists** in the F1 expression used in the proof architecture.

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